```
adyanth@debian: ~/PES1UG24CS036-pes-vcs
a string of length 511 [-Wstringop-truncation]
 169 |         strncpy(e->path, path, sizeof(e->path) - 1);
     |         ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
adyanth@debian:~/PES1UG24CS036-pes-vcs$ export PES_AUTHOR="Your Name <PESXUG24CS
042>"
adyanth@debian:~/PES1UG24CS036-pes-vcs$ export PES_AUTHOR="Your Name <PES1UG24CS
036>"
adyanth@debian:~/PES1UG24CS036-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
adyanth@debian:~/PES1UG24CS036-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
adyanth@debian:~/PES1UG24CS036-pes-vcs$
```

```
All Phase 1 tests passed.
adyanth@debian:~/PES1UG24CS036-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
adyanth@debian:~/PES1UG24CS036-pes-vcs$
```

adyanth@debian: ~/PES1UG24CS036-pes-vcs

```
adyanth@debian:~/PES1UG24CS036-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b
9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
adyanth@debian:~/PES1UG24CS036-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
adyanth@debian:~/PES1UG24CS036-pes-vcs$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
adyanth@debian:~/PES1UG24CS036-pes-vcs$
```

```
adyanth@debian:~/PES1UG24CS036-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
cat .pes/index    # Human-readable text format
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  object.c
  untracked:  tree.h
  untracked:  commit.c
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  test_objects
  untracked:  tree.c
  untracked:  README.md
  untracked:  Makefile
  untracked:  pes.h
  untracked:  test_tree
  untracked:  test_objects.c
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  REPORT.md
  untracked:  index.h
  untracked:  commit.h

100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1777545973 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1777545973 6 file2.txt
adyanth@debian:~/PES1UG24CS036-pes-vcs$
```

```
adyanth@debian:~/PES1UG24CS036-pes-vcs$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 15de1fa1c38e... Initial commit
Committed: c790b2f9cb34... Add world
Committed: b70b20475c6f... Add farewell
commit b70b20475c6fec65a63e8e7f4bf6b29d07ae930aaec95fc68257ea81e810fbdb
Author: Your Name <PES1UG24CS036>
Date:   1777546047

    Add farewell

commit c790b2f9cb34594476494c52b9438c3573531d6667a153b1a8cf8d47edcbb167
Author: Your Name <PES1UG24CS036>
Date:   1777546047

    Add worldell

commit 15de1fa1c38e3f5481730995840ffdbedf825faf2c8f1a2b37ca3def09382ed5
Author: Your Name <PES1UG24CS036>
Date:   1777546047

    Initial commit

adyanth@debian:~/PES1UG24CS036-pes-vcs$
```

```
adyanth@debian:~/PES1UG24CS036-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/15/de1fa1c38e3f5481730995840ffdbedf825faf2c8f1a2b37ca3def09382ed5
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/30/1c0972d90ecf51963f0b154ea2698c52960ba76cfccafadd7f4169ef2597f0
.pes/objects/58/51a75527e7cfe8b2b4635ce41983b5c31e61cfa6bf99bbd2a4f9602fe22a2c
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/78/2cf7db60a833c87e4ea9953802d0c8973d00bd3e7c594a9b10a78e3550ec16
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/b7/0b20475c6fec65a63e8e7f4bf6b29d07ae930aaec95fc68257ea81e810fbdb
.pes/objects/c7/90b2f9cb34594476494c52b9438c3573531d6667a153b1a8cf8d47edcbb167
.pes/objects/c9/6f2fc25394979be592d5cf80c5d78ec792ff13dde17f838449fbcdfbb7df0d
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

```
adyanth@debian:~/PES1UG24CS036-pes-vcs$ cat .pes/refs/heads/main
782cf7db60a833c87e4ea9953802d0c8973d00bd3e7c594a9b10a78e3550ec16
adyanth@debian:~/PES1UG24CS036-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

adyanth@debian: ~/PES1UG24CS036-pes-vcs

```
adyanth@debian:~/PES1UG24CS036-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: a555f9877560... Initial commit

Log after first commit:
commit a555f9877560dee7a78e2883257e3d4efad770112c95c32d01fbe77f49c8f089
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Initial commit


--- Second Commit ---
Committed: e5d31e5a91a5... Update file.txt

--- Third Commit ---
Committed: 95fb0fc534c0... Add farewell

--- Full History ---
commit 95fb0fc534c06bd0892b4f8de75e54fa4e871052ce9142b7a0d4c6eab01f42d4
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Add farewell

commit e5d31e5a91a5b3bf50e342ba2ca461ccfd1134704a43562f04fecfb10e153d01
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Update file.txt
```

```
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: a555f9877560... Initial commit

Log after first commit:
commit a555f9877560dee7a78e2883257e3d4efad770112c95c32d01fbe77f49c8f089
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Initial commit


--- Second Commit ---
Committed: e5d31e5a91a5... Update file.txt

--- Third Commit ---
Committed: 95fb0fc534c0... Add farewell

--- Full History ---
commit 95fb0fc534c06bd0892b4f8de75e54fa4e871052ce9142b7a0d4c6eab01f42d4
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Add farewell

commit e5d31e5a91a5b3bf50e342ba2ca461ccfd1134704a43562f04fecfb10e153d01
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Update file.txt

commit a555f9877560dee7a78e2883257e3d4efad770112c95c32d01fbe77f49c8f089
Author: Your Name <PES1UG24CS036>
Date:   1777546217

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
95fb0fc534c06bd0892b4f8de75e54fa4e871052ce9142b7a0d4c6eab01f42d4

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd90b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef6b4c7c7e0d6695797533f50d
.pes/objects/95/fb0fc534c06bd0892b4f8de75e54fa4e871052ce9142b7a0d4c6eab01f42d4
.pes/objects/a5/55f9877560dee7a78e2883257e3d4efad770112c95c32d01fbe77f49c8f089
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa80c09635c5895ef7e08f7fa1974d901ce282f30e00de0ccd92
.pes/objects/db/7733c25d2d137b7574bed6c5542b62bf48baf6ea3829f61f75b0d10d5358f9
.pes/objects/db/87a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e5/d31e5a91a5b3bf50e342ba2ca461ccfd1134704a43562f04fecfb10e153d01

=== All integration tests completed ===
adyanth@debian:~/PES1UG24CS036-pes-vcs$
```